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The effect of glycidyl methacrylate and styrene comonomers on compatibility, physical-mechanical properties, and swelling behavior of SBR/NBR blends

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Abstract

The effect of functionalized styrene-butadiene rubber (SBR) with glycidyl methacrylate (GMA) monomers, in the presence of styrene comonomers (SBR-g-GMA-co-St) as a compatibilizer on physical-mechanical, morphological, and swelling behavior of SBR/nitrile butadiene rubber (NBR) blends were studied. It was proved that the compatibilizer made through dicumyl peroxide-induced reaction which possessed 1.5 phr GMA in addition to 1.5 phr styrene comonomers showed higher efficiency of grafting. Possible in situ reactions between SBR-g-GMA-co-St and NBR component also were evaluated with an attenuated total reflectance mode of Fourier-transform infrared spectroscopy. SBR/NBR blend containing compatibilizer with a middle value of grafting demonstrated the most reactions between components. Dynamic mechanical thermal analysis results illustrated that the presence of SBR-g-GMA-co-St caused a significant improvement in compatibility of two components. This was verified with the scanning electron microscope pictures in which a smoother surface of the sample was clear. The enhancement in microstructure led to an increase in tensile strength, elongation at break, and storage modulus. Moreover, the increase in the intermolecular cross-links and made interactions considerably affected blends' swelling behavior in both hydrocarbon solvents (carbon tetrachloride and chloroform).

KEYWORDS

blending, compatibilization, swelling, vulcanization

1 | INTRODUCTION

Polymer blending is an effective and economic approach in order to fabricate novel materials with appropriate properties.¹ Most of polymer blends are immiscible because of their high molecular weight and the low entropy of mixing and blending of the most of polymers results in improper physical and mechanical properties.² As a result, using a suitable compatibilizer agent is useful to achieve outstanding properties.³ One of the major

methods to compatibilize components is addition of the block, random copolymer segments, or grafts, which can decrease surface tension and create better homogeneity in a system by increasing interfacial adhesion between blends components.⁴ The compatibilizers could act as bridges between phases and transfer the imposed stress since it makes a compatible region at interface of components.⁵ Styrene-butadiene rubber (SBR) has suitable processability and acceptable mechanical properties in addition to the ozone resistance, weatherability, and high

fatigue life.⁶ However, it swells vigorously in the vicinity of oils; therefore, blending of the SBR with nitrile butadiene rubber (NBR) makes it resistant to nonpolar oils and solvents.^{7,8} It is predicted that the styrene and nitrile groups in SBR and NBR would act as an electron donor-acceptor agent and allow compatibility of this blend,¹ but the experimental observations illustrated that SBR and NBR are incompatible.^{9–11} In order to optimize the properties of SBR/NBR blends utilizing a proper compatibilizer would be undoubtedly useful.^{12,13} Unsaturated maleic anhydride (MAH), cis-butenedioic acid, carboxylic acid esters, acrylic acid, and its esters like glycidyl methacrylate (GMA) are most frequently used for increasing the tendency of components; consequently, the increase the compatibility of the system.^{14,15} Investigation on the effect of chemical essence in MAH and GMA as most frequency compatibilizer in polyolefins industry demonstrated that double bond of MAH monomers had much lower reactivity than that of GMA monomers as a result of the higher electron tendency of carbonyl groups of MAH monomers compared with GMA monomers.¹⁶ In addition, to higher interaction interest of the end groups in GMA monomers and cyanide group in NBR component.¹⁷ Hence, GMA was selected instead of MAH. Furthermore, to improve the grafting reaction, adding a vinyl monomer like styrene seems to be helpful.¹⁸ Employing styrene monomers could increase the grafting yield by reducing undesirable reactions such as cross-linking and chain scission of SBR.^{19,20} Moreover, interfacial adhesion in elastomeric blends can be reinforced by cocuring of different phases.²¹ However, a major drawback of covulcanization of SBR and NBR is the difference between their cure rate index (CRI) due to the discrepancy in polarity and inherent diversity of these components.

In order to improve the efficiency of the grafting reaction of GMA onto SBR, analytical optimization was used. Response surface methodology (RSM) is one of the most practical multivariate techniques in analytical optimization that is based on desirability function and artificial intelligence.^{22,23} The intrinsic of this method is to fit tentative models on the experimental information gained from the experimental designs. Suitable design of an experiment of grafting reaction is one of the primary stages to reach the reliable data. Central composite design (CCD) is used for modeling the grafting reaction between GMA and SBR in presence of styrene comonomers and gelation of SBR. The advantage of this design is an access to the response of a method with the least number of samples. CCD base is numbering the parameters that involved in the experiment by three levels: least, middle, and maximum amount of variable parameter (α^- , α_{mean} , α^+).^{24,25}

This study was interested in increasing the compatibility of the SBR/NBR blends by the formation of intermolecular cross-links and interactions between graft copolymer SBR-g-GMA-co-styrene and covulcanization of phases. The compatibilizer was prepared into an internal mixer and in order to investigate the factors affecting the grafting reaction, the amount of GMA and styrene/GMA ratio were chosen as variable parameters, and to optimize the variable parameters, experimental design using RSM on calibrated Fourier-transform infrared (FTIR) spectroscopy was employed. The effect of prepared compatibilizers on rheological behavior of the SBR/NBR blends containing 10 phr of the compatibilizer was investigated using an oscillating disc rheometer (ODR). The compatibility of the SBR/NBR blends with and without compatibilizer was evaluated with dynamic mechanical thermal analysis (DMTA) and to investigate possibility of in situ compatibilization between compatibilizers and NBR component FTIR (attenuated total reflectance [ATR] mode) was used. The effect of compatibilizers on physical and mechanical properties such as tensile test, hardness, and swelling were examined in carbon tetrachloride and chloroform.

2 | EXPERIMENTAL

2.1 | Materials

Acrylonitrile butadiene rubber (NBR 3450) with 32% of the acrylonitrile, Mooney viscosity (ML 1 + 4100°C = 52 MU), and density = 1.17 g/cm³ was purchased from Bayer AG, Germany. Styrene-butadiene rubber (SBR 1502) with approximate 23% styrene, Mooney viscosity (ML 1 + 4100°C = 55 MU), and density of 0.93 g/cm³ was purchased from Bandar-Imam Petrochemical Co. Ltd. Dicumyl peroxide (DCP), styrene monomer (99% purity), and GMA with 97% purity having density values of 1.56, 1.08, and 0.90 g/cm³, respectively, were provided by Merck Co. Ltd. Chlorinated hydrocarbons solvents (carbon tetrachloride and chloroform) with molar volume values of 100 and 80.7 cm³/mol and having density of 1.59 and 1.49 g/cm³, respectively, were supplied by Dr.Mojallali Co. N-cyclohexyl-2-benzothiazole sulfonamide (CBS) and tetramethyl thiuram disulfide (TMTD) were purchased from Anchor chemical Co. Zinc oxide (ZnO), stearic acid (StAc), sulfur, N330, and processing oil were of commercial grades.

2.2 | Grafting procedures

Grafting of GMA onto SBR in the presence of styrene monomer was carried out through free-radical grafting

mechanism on Brabender PL2200, 60 cc. First SBR was dried in oven at 80°C for 6 hours, then it was added into the Brabender chamber to melt at 120°C with rotor speed of 50 rpm. After 2 minutes, styrene comonomer was added into the chamber, afterward GMA monomer and DCP were added after 1 and 2 minutes, respectively, as schematically shown in Figure 1. To prevent continuous reaction after discharging them from the chamber, the products were immediately plunged in cold water.²⁶

2.2.1 | Titration and determination of grafting ratio

The samples obtained from internal mixer were weighted and dissolved into toluene (150 mL) at 65°C about 4 hours; afterward, precipitated with extra amount of water (250 mL), after adding water, three phases were formed as follows: (a) The main phase containing SBR grafted with GMA, styrene, and unreacted DCP, (b) aqueous phase containing unreacted GMA and styrene monomers, and (c) gel containing cross-linked SBR.²⁷ The organic phase was separated through titration process with KOH and ethanol solution with presence of 0.1 g:N sodium hydroxide thymol blue as an indicator. Before appearance of change in color of the system (creamy to blue), KOH was being added consistently. Then, it was titrated with hydrochloric acid (until the color back to the initial color).²⁸ The mixture was kept at 60°C for 3 hours in order to complete the reaction between HCl with grafted GMA. The yield of GMA grafting was implied by creating a calibration curve of FTIR with titration results. For grafted samples, a peak appears in the FTIR spectra at around 1730 cm⁻¹, which is corresponds to stretching of carbonyl groups of grafted GMA (—C=O),²⁹ while the transmittance peak at 700 cm⁻¹ attributed to the stretching of styrene group in SBR. The ratio of the peak area of GMA carbonyl to the styrene group of SBR (I_{1730}/I_{700}) was associated to the absolute GMA grafting yield measured by the

titration method.³⁰ To calculate the GMA grafting ratio, samples that obtained from CCD with constant concentration of DCP relative to SBR content (0.5/100 phr), variable GMA content of 1, 1.5, and 2 phr, and variable styrene/GMA ratio of 0.5, 1, and 1.5 were measured by FTIR and the calibration curve indicating the relationship between I_{1730}/I_{700} and grafting yield was gained. Nine different samples that had been obtained from RSM, with different initial GMA content and styrene/GMA ratio were prepared to make a calibration curve.

2.2.2 | Gel content measurement

The gel amount was determined from Soxhlet extraction. At first, 1 g of the grafted sample was refluxed with toluene for 24 hours at 180°C and the obtained material was dried in a vacuum oven for 6 hours at 60°C. The dried sample was weighted and the amount of the gel was calculated according to Equation 1.

$$\% \text{Gel} = \frac{W_1 - W_2}{W_1} \times 100, \quad (1)$$

where W_1 and W_2 indicate weight of the sample before and after the Soxhlet extraction process.

2.3 | Compatibility procedures

Mentioned blends could be compatibilized by three procedures. First, creation of cross-links between NBR and SBR domains during the cocure process. Second one is that interfacial compatibilization of oxarine end groups in SBR-g-GMA and cyanide groups in NBR makes oxazoline structure. Last path is to reduce the interfacial tension of phases by the intermolecular hydrogen bonding formation between compatibilizer and NBR phase. FTIR spectra were used to determine the reactive

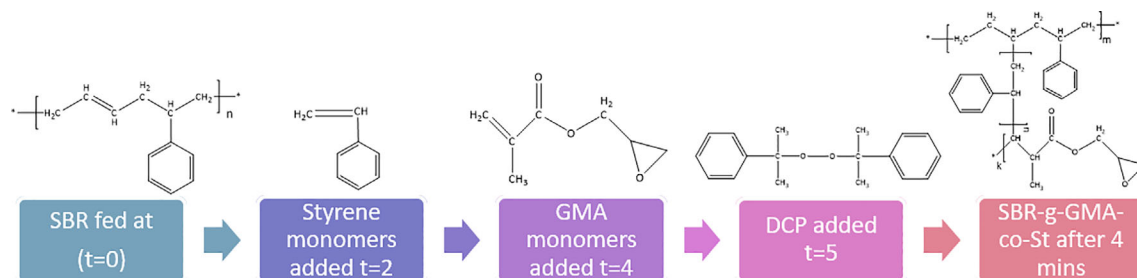


FIGURE 1 Time schedule applied in DCP-induced grafting of GMA monomers onto SBR in presence of styrene. DCP, dicumyl peroxide; GMA, glycidyl methacrylate; SBR, styrene-butadiene rubber [Color figure can be viewed at wileyonlinelibrary.com]

compatibilization of oxarine end groups and cyanide groups, that will be discuss in the Result section.

2.4 | Preparation and vulcanization of blends

The rubber blends were mixed with 10 phr of the compatibilizer (SBR-g-GMA-co-St) on a two-roll mill of 480 mm diameters and 300 mm gap at room temperature. The speed of the slow roller was 30 rpm with 1:1.5 friction, according to ASTM D3182 under control of rolls' temperature. Compatibilizer (SBR-g-GMA-co-St), carbon black N330, CBS, TMTD, ZnO, stearic acid, sulfur, 2,2,4-trimethyl-1,2-dihydroquinoline, and processing oil were used for the compounding process. Formulations of compounded blends have shown in Table 1. The fabricated compounds were vulcanized through hot press at the temperature of 160°C with compression of 4 MPa according to cure times (t_{90}) as given from rheometer data. Blends were entitled based on the amount of SBR in the blend and produced compatibilizers (R1 to R9) which were defined in Table 2 (eg, S₂₀/R1 indicates SBR/NBR [20/80] with 10 phr of R1 compatibilizer).

2.5 | Rheological measurements

Cure rheometric characteristics (M_H , M_L , T_2 , T_{90} , and CRI) were evaluated using by a Monsanto rheometer ODR 100S according to ASTM D2084-17. About 7 g of compounded blends and components of blends were tested at the temperature of 160°C to assess the effect of compatibilizer and the cure system to rheometric characteristics.

2.6 | Fourier-transform infrared spectroscopy

Characterization of compatibilization was carried out using BOMEM-Canada-Model 650 spectrometer. About 1 g of compatibilizers that designed through CCD (R1, R4, and R5) were chopped to small part to increase the contact surface, then, were Soxhlet with acetone at the temperature of 40°C for 6 hours to separate the unreacted GMA and styrene monomers from compatibilizers. Afterward, the remained samples were dried in vacuum oven for 2 hours. The purified compatibilizers from GMA and styrene monomers were Soxhlet again with toluene at 150°C for 24 hours to separate cross-linked SBR from compatibilizer. The solution of SBR-g-GMA-co-St in toluene was collected by glass syringe and minimum five droplets of solution were seeped on KBr tablet. After evaporation of toluene, the tablets were compressed and FTIR spectroscopy was employed. Furthermore, possible reaction between epoxy end groups of SBR-g-GMA-co-St with NBR component, surface of R1 compatibilizer, cured NBR, and 50/50/10 (SBR/NBR/SBR-g-GMA-co-St) were evaluated using ATR mode with the same spectrometer.

2.7 | Dynamic mechanical thermal analysis

DMTA was measured using TTDMA (TRION, England). The vulcanized NBR, SBR, and 50/50 composition ratio of their blends with three different produced compatibilizers in dimensions of 10.5 × 1.5 × 4.5 mm were tested in tension mode with the frequency of 1 Hz and

Ingredients (phr)	Blend				
	S ₀	S ₂₀	S ₅₀	S ₈₀	S ₁₀₀
SBR	0	20	50	80	100
NBR	100	80	50	20	0
Compatibilizer	0	0 and 10	0 and 10	0 and 10	0
Zinc oxide	5	5	5	5	5
Stearic acid	2	2	2	2	2
Sulfur	1.2	1.2	1.2	1.2	1.2
CBS	1	1	1	1	1
TMTD	0.2	0.2	0.2	0.2	0.2
TMQ	1	1	1	1	1
Processing oil	6	6	6	6	6
Carbon black	30	30	30	30	30

TABLE 1 Exposition of the blends

Abbreviations: CBS, N-cyclohexyl-2-benzothiazole sulfonamide; NBR, nitrile butadiene rubber; SBR, styrene-butadiene rubber; TMQ, 2,2,4-trimethyl-1,2-dihydroquinoline; TMTD, tetramethyl thiuram disulfide.

TABLE 2 Grafting characteristics and gel content of SBR

Sample name	GMA content (phr)	Styrene/GMA	Grafting yield (%wt)	Amount of gel (%)
R1	1.5	1.5	2.8	48
R2	1.0	1.5	1.2	59
R3	2.0	1.5	1.6	43
R4	1.5	1.0	3.3	61
R5	1.5	0.5	1.4	68
R6	2.0	1.0	3.1	57
R7	1.0	1.0	2.8	60
R8	1.0	0.5	1.3	72
R9	2.0	0.5	1.6	65

Abbreviations: GMA, glycidyl methacrylate; SBR, styrene-butadiene rubber.

temperature range of -60°C to 25°C with the heating rate of $3^{\circ}\text{C}/\text{min}$.

2.8 | Tensile test

The tensile strength, modulus (M100), and elongation at break were determined with a Galdabini Sun2500 (Italy). Three dumbbell-shaped specimens were cut from the vulcanized sheets with suitable punch dies with 4 mm width, neck length of 20 mm, and the thickness of 2 mm according to ISO 12244. The samples were stretched with the speed of 50 mm/min according to ASTM D412 for the determination of tensile strength properties.

2.9 | Hardness test

One of the essential tests in physical and mechanical properties of the elastomer blends is hardness test, high degree of hardness. In addition, the reference material depends on other factors such as the amount of carbon black, cross-link density, and so on.³¹ The hardness test was carried out on the vulcanized samples 8 mm thick were tested with a Shore A durometer according to DIN 53505 using a Frank Shore durometer (Germany).

2.10 | Scanning electron microscopy

The morphology of SBR/NBR blends (50/50) with and without compatibilizers was analyzed using a scanning electron microscope (SEM) so as to study the effect of different sort of compatibilizers on the compatibility and fracture surface of the blends. The micrograph was taken using AIS-2100 SEM supplied by SERON company

through the gold-coated fracture surface of tensile specimens.

2.11 | Cross-link density and swelling measurements

In order to study the swelling behavior of the blends, carbon tetrachloride and chloroform were chosen as a polar and nonpolar solvents. About 1 g ($7 \times 7 \times 2$ mm) of the vulcanized sheets were cut, weighted, and plunged into 100 mL of each solvent. The samples were periodically removed, their surfaces cleaned, weighted, and placed back into the solvents. This process continued until the weight of samples became unchanged.³² All the swelling experiments were performed at ambient temperature and regardless of weighting time, the samples were kept in dark. The minimum of two samples were averaged and demonstrated as the results.

The swelling data were used to determine the molecular weight between two cross-links, M_c (g/mol), utilizing Flory-Rehner Equation³³:

$$M_c = \frac{-\rho_p V \phi^{\frac{1}{3}}}{(\ln(1-\phi) + \phi + \chi \phi^2)}, \quad (2)$$

$V(\text{CCl}_4)$ 100 cm^3/mol ; $V(\text{CHCl}_3)$ 80.7 cm^3/mol ; $\rho_s(\text{CCl}_4)$ 1.54 g/cm^3 ; $\rho_s(\text{CHCl}_3)$ 1.49 g/cm^3 where ρ_p is the density of blend components, V is the molar volume of the solvent, ϕ is the volume fraction of swollen rubber and can be obtained from the masses and densities of rubber samples and solvents, and χ is the interaction parameter of blend components that can be gained from Equation 4:

$$\chi = \frac{\frac{d\phi}{dT} \left(\left(\frac{\phi}{1-\phi} \right) + N \ln(1-\phi) + N\phi \right)}{2\phi \left(\frac{d\phi}{dT} \right) - \phi^2 N \frac{d\phi}{dT} - \frac{\phi^2}{T}}, \quad (3)$$

where $\frac{d\phi}{dT}$ is volume fraction relevant to temperature, N is parameter dependent to volume fraction and can be calculated from Equation 5:

$$N = \frac{\frac{\phi^3}{3} - \frac{2}{3}}{\phi^{\frac{1}{3}} - 2\frac{\phi}{3}}. \quad (4)$$

From the values of M_c , the cross-link density, ν (g/cm³), has been determined as:

$$\nu = \frac{1}{(2M_c)}. \quad (5)$$

3 | RESULT AND DISCUSSION

3.1 | Characterization of grafting

The amount of grafted GMA on SBR chains was determined from the area ratio of FTIR peaks at 1730 and 700 cm⁻¹ and was calibrated with data which had been obtained from the titration method. Figure 2A shows FTIR spectra of R1, R4, and R5 samples which were chosen through CCD. This design was used for all produced compatibilizers. Table 2 represents the GMA grafted and gel content of SBR for all samples.

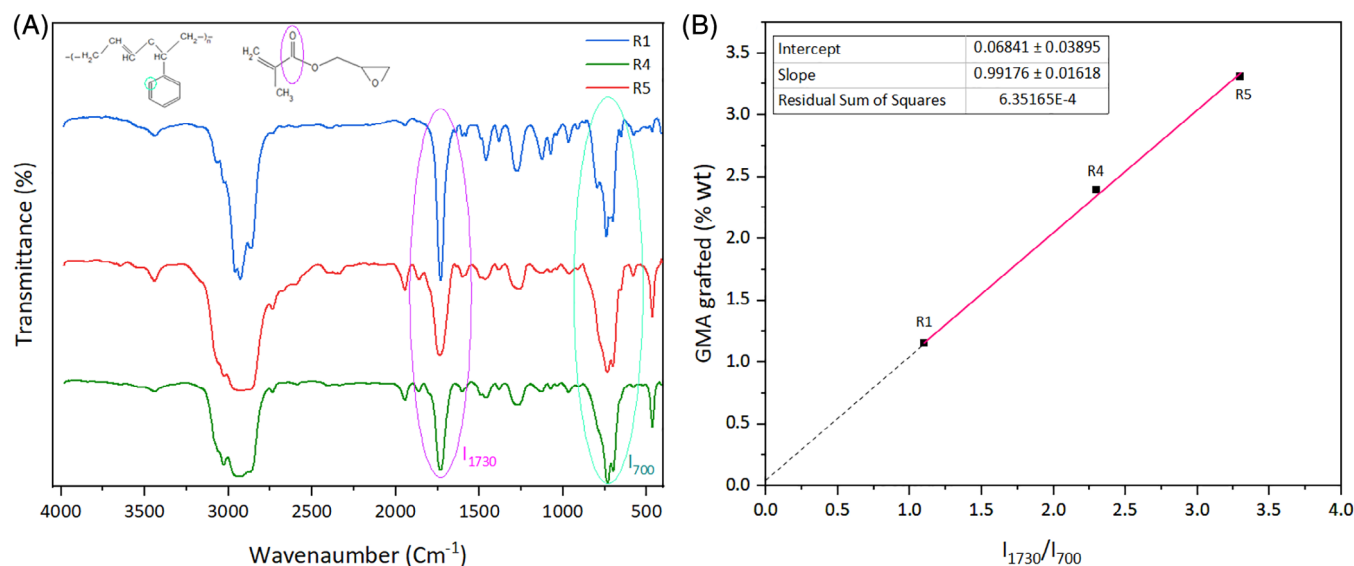


FIGURE 2 A, FTIR spectra samples with DCP content of 0.5 phr and variable GMA contents of 1, 1.5, and 2 phr and styrene/GMA ratio of 0.5, 1, and 1.5 phr. B, FTIR calibration curve of GMA grafting on SBR. DCP, dicumyl peroxide; FTIR, Fourier transform infrared; GMA, glycidyl methacrylate; SBR, styrene-butadiene rubber [Color figure can be viewed at wileyonlinelibrary.com]

3.1.1 | Effect of variable parameters on grafting yield

In order to study and optimize the effect of variable parameters (GMA content and styrene/GMA ratio), the CCD was used. As it can be seen in Figure 3A, the increasing amount of GMA did not raise the grafting yield, and this probably happened because of the separation of GMA monomers from rubbery phase and formation of distinct GMA phase; therefore, reduction in the contingency of GMA radical reaction with the double bond of SBR. Moreover, the styrene/GMA ratio plays a crucial role in grafting yield determination. The highest grafting yield amount was reached when the styrene/GMA ratio of 1 had been used.

3.1.2 | Effect of variable parameters on amount of gel

With the purpose to reduce the amount of gel, the same design of experiment as in previous section was used. As it is observed in Figure 3B, the increase in both variable parameters (GMA content and styrene/GMA ratio) indicates decline in gel content that is due to the competitive reaction between chains cross-linking with homopolymerization of styrene and GMA monomers. It should be noted that, both homopolymerization reactions have the advantage of free radical absorption compared with the chain cross-linking reaction which because of free movement of styrene and GMA monomers compared to

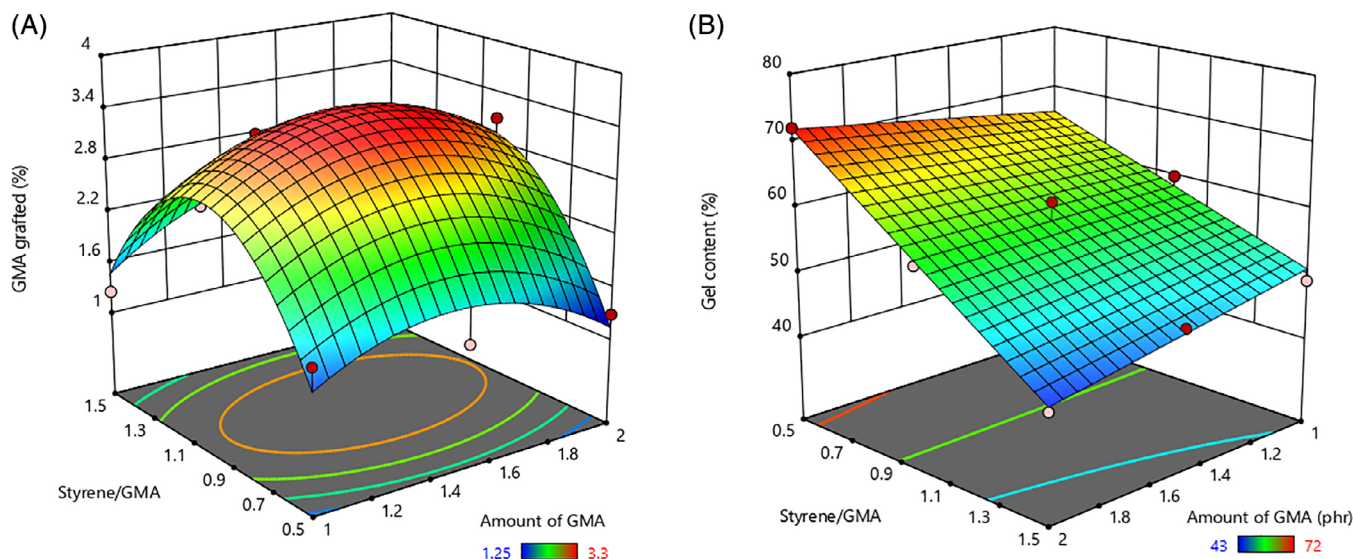


FIGURE 3 Effect of variable parameters, A, GMA graft yield on SBR and B, % gel content. GMA, glycidyl methacrylate; SBR, styrene-butadiene rubber [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 3 Cure characteristics of the blend

Samples code	Torque (dN·m)		Time (min', s")		Rate of cure (min ⁻¹) CRI
	M_L	M_H	T_2	T_{90}	
S ₀	11.5	49.1	4.3	9.1	20.9
S ₂₀	12.6	49.4	4.1	8.2	25.1
S ₂₀ /R1	11.4	49.9	3.5	7.4	25.6
S ₅₀	13.3	46.3	3.3	7.1	26.7
S ₅₀ /R1	11.5	49.2	3.2	6.5	27.9
S ₅₀ /R4	13.4	48.3	3.1	6.5	28.9
S ₅₀ /R5	13.2	47.8	3.3	7.1	27.1
S ₈₀	13.1	50.1	2.4	6.1	27.1
S ₈₀ /R1	12.8	51.6	2.4	5.4	33.2
S ₁₀₀	12.9	53.1	2.3	5.1	34.9

Abbreviation: CRI, cure index rate.

the movement of polymer chains in the reaction environment. Furthermore, in order to avoid stress concentration that would cause physical and mechanical fragility, gel-free SBR-g-GMA-co-St was purified by Soxhlet extraction and then used as the compatibilizer.

3.2 | Rheological characteristics

Rheological characteristics of SBR, NBR, and all blends' with and without SBR-g-GMA-co-St in three levels of grafting yield (maximum, medium, and minimum) were carried out and summarized in Table 3. It can be seen that the minimum torques M_L decreased in the presence of compatibilizer and the lowest value of it was obtained

when medium grafting yield compatibilizer (R1) had been used. This demonstrated that R1 compatibilizer possessed more efficiency to develop processability into the blend compared to R4 and R5 compatibilizers. Also, attendance of compatibilizers increased in the maximum torque, M_H , and the highest value acquired when R1 compatibilizer was used.

The scorch time t_2 and the cure time t_{90} of the blends decreased with the presence of compatibilizer, and the lowest of them was attained when R1 compatibilizer had been used. The results of rheological characteristics of SBR/NBR (50/50) with three level of grafting yield showed that average grafting yield (R1) was created with the least value of t_2 , t_{90} , and M_L and maximum value of M_H . As it had the best rheological properties, it has been

selected for all of the blend ratios, for which data are shown in Table 3. It could be seen that the t_2 and t_{90} are shorter with the presence of SBR-g-GMA-co-St. This

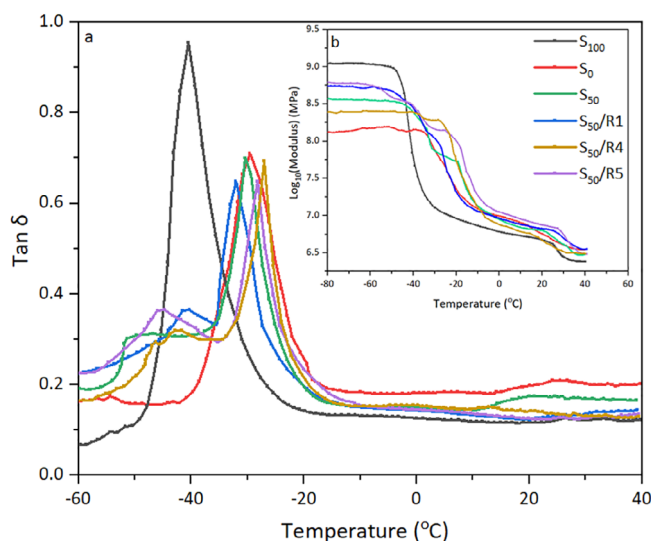


FIGURE 4 DMTA analysis of NBR, SBR, and NBR/SBR (50/50) with R1, R4, and R5 compatibilizers and NBR/SBR (50/50) without compatibilizer. A, Tan delta vs temperature. B, Storage modulus against temperature. DMTA, dynamic mechanical thermal analysis; NBR, nitrile butadiene rubber; SBR, styrene-butadiene rubber [Color figure can be viewed at wileyonlinelibrary.com]

situation could occur for two reasons: (a) activation of an adjacent double bond by the epoxide group³⁴ and (b) lower t_2 and t_{90} of SBR-g-GMA-co-St compared with SBR. On the other hand, SBR/NBR blend exhibited lower M_L and higher M_H with SBR-g-GMA-co-St. This indicates better processability and better interaction between components of the blend, and probably higher cross-link density.

3.3 | Compatibility investigation

Compatibility investigation was conducted on measurements of glass transition temperature (T_g) and FTIR spectra (ATR mode) to determine the reactive compatibilization. Figure 4 shows results obtained from DMTA test for SBR/NBR (50/50) blend, with R1, R4, and R5 compatibilizers compared to SBR/NBR (50/50) blend without compatibilizer and components of the blend. As it can be seen, two distinguishable peaks appeared in the 50/50 blends which present immiscible behavior of two components of the blend. As the compatibility of components improved, these peaks subsequently got closer together and the weight of the peaks increased. This was in light of shrinkage of the components' domains due to the decrease in interfacial tension between components in addition to more interference of component's chains which led to increase in the weight of the peaks.

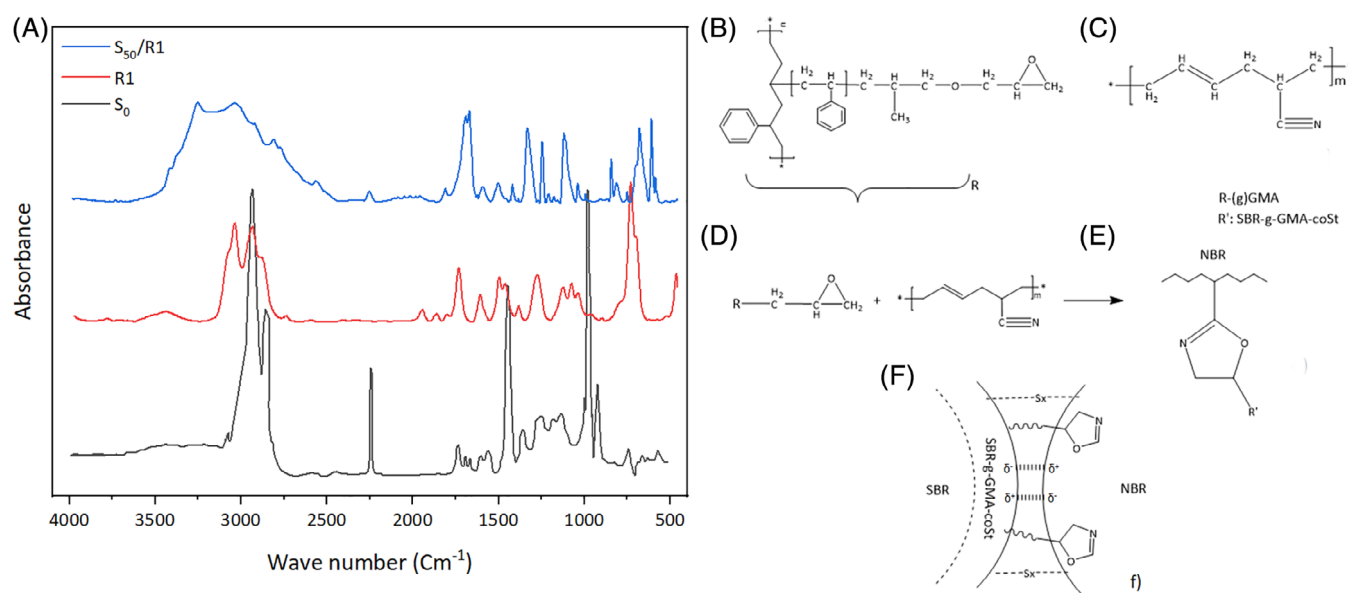


FIGURE 5 A, FTIR absorption spectra curve (ATR) of NBR, R1 compatibilizer, and SBR/NBR (50/50) blend with 10 phr of R1 compatibilizer and schematic of grafting GMA onto SBR: B, SBR-g-GMA-co-St structure, C, NBR structure, D, Reaction of epoxy end-groups of SBR-g-GMA-co-St with NBR and creating compatibility by developing of oxazoline structure, and E, Possible intermolecular interactions and cross-links between two components of the blend. ATR, attenuated total reflectance; FTIR, Fourier transform infrared; GMA, glycidyl methacrylate; NBR, nitrile butadiene rubber; SBR, styrene-butadiene rubber [Color figure can be viewed at wileyonlinelibrary.com]

Furthermore, the vibration of first component chains during the transition temperature causes the vibration of another component chains; in a reverse logic, one component would have held the others to vibrate freely. Consequently, two tan delta peaks approach toward each other as a result of the increase in the compatibility.³⁵ The sample having R1 compatibilizer showed a closer tan delta peaks and more area under the peaks rather than other samples, which means better dispersion and more interactions between phases' domains.^{36,37} It is worthy to say that an increase in epoxy groups in the R4 compatibilizer might have raised the viscosity of compatibilizer which probably inhibited the domains' separation; subsequently, reducing the compatibility compared to the R1 compatibilizer.

Moreover, the characteristic peaks in the FTIR analysis showed absorption peaks attributed to carbonyl (C=O) stretching vibration of GMA grafted on SBR chains. The absorption peak at about 2240 cm^{-1} corresponded to cyanide groups of NBR (C≡N).^{3,38}

As it can be seen in SBR/NBR(50/50) blend with 10 phr of SBR-g-GMA-co-St FTIR spectra, the absorption peak at about 1690 cm^{-1} is attributed to (C=N) and peak at about 700 cm^{-1} is attributed to (C—O) of oxazoline structure, which verifies the occurrence of interfacial in situ compatibilization by development of oxazoline structure. Furthermore, oxirane groups and cyanide group from NBR can enhance the compatibility of phases through intermolecular hydrogen bonding.³⁹ Figure 5E shows compatibilization of mentioned blend trough intermolecular cross-links and hydrogen bonds and develop of oxazoline structure.

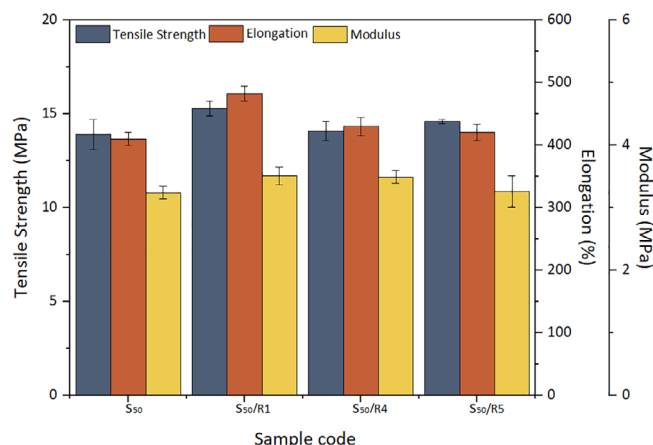


FIGURE 6 Tensile properties of NBR/SBR (50/50) blends without and with 10 phr of R1, R4, and R5 compatibilizer. NBR, nitrile butadiene rubber; SBR, styrene-butadiene rubber [Color figure can be viewed at wileyonlinelibrary.com]

3.4 | Physical mechanical properties

3.4.1 | Tensile properties

To determinate the physical and mechanical properties, the tensile strength test was carried out. The characterization of the tensile properties' of 50/50 blends without and with R1, R4, and R5 compatibilizers showed in Figure 6. The sample containing R1 compatibilizer showed the higher tensile strength value compared to the other samples which could be due to the better distribution of domains; consequently, an increase in the aspect ratio of domains and better dimensional stability of the sample against tension.³⁹ Elongation at break demonstrated the same trend, the E_b of sample containing the R1 compatibilizer increased by 70% compared to the uncompatibilized blend, that could be the result of an increase in cross-links' density and higher interaction among the phases. The effect of compatibilizer on Young's modulus of the sample possessing R5 compatibilizer was almost negligible due to the shortage of an epoxy groups onto the surface of R5 compatibilizer. However, the samples which contained R1 and R4 compatibilizers showed detectable increase in the modulus comparing to uncompatibilized blend. Occurrence of hydrogen bonds between double bond of NBR and epoxy groups of the compatibilizer, which prevent the development of sulfide links between intradomains, probably led to the isolation of these double bonds which subsequently increased cross-links between the phases and raised the modulus.⁴⁰ Furthermore, the effect of blend compositions at presence of R1 compatibilizer; as an optimized state, was illustrated in Figure 7. As it can be seen in Figure 7A, addition of NBR resulted in decrease in the tensile strength at the S_{80} which mostly occurred because of incompatibility of blend.¹² However, after appearance of the state at which NBR became the dominant phase, tensile strength increased as NBR increased due to higher stability of NBR associated with polarity of NBR molecules. Figure 7B shows the effect of composition on the elongation at break. The results indicated that pure NBR and SBR had higher E_b compared to their blends. This negative effect might be because of shortage in the intramolecular links in the uncompatibilized blend⁴¹ and it was confirmed by compatibilized blends' result. The results of tensile modulus illustrated an increase in NBR content caused gradual increase in modulus of the blend (Figure 7C). The results of tensile modulus were in agreement with storage modulus of the DMTA test. However, there might be a slight difference between the values due to different instinct of the tests, albeit the values followed same trend. This observation indicated that polarity of system had important effect on modulus of the system by

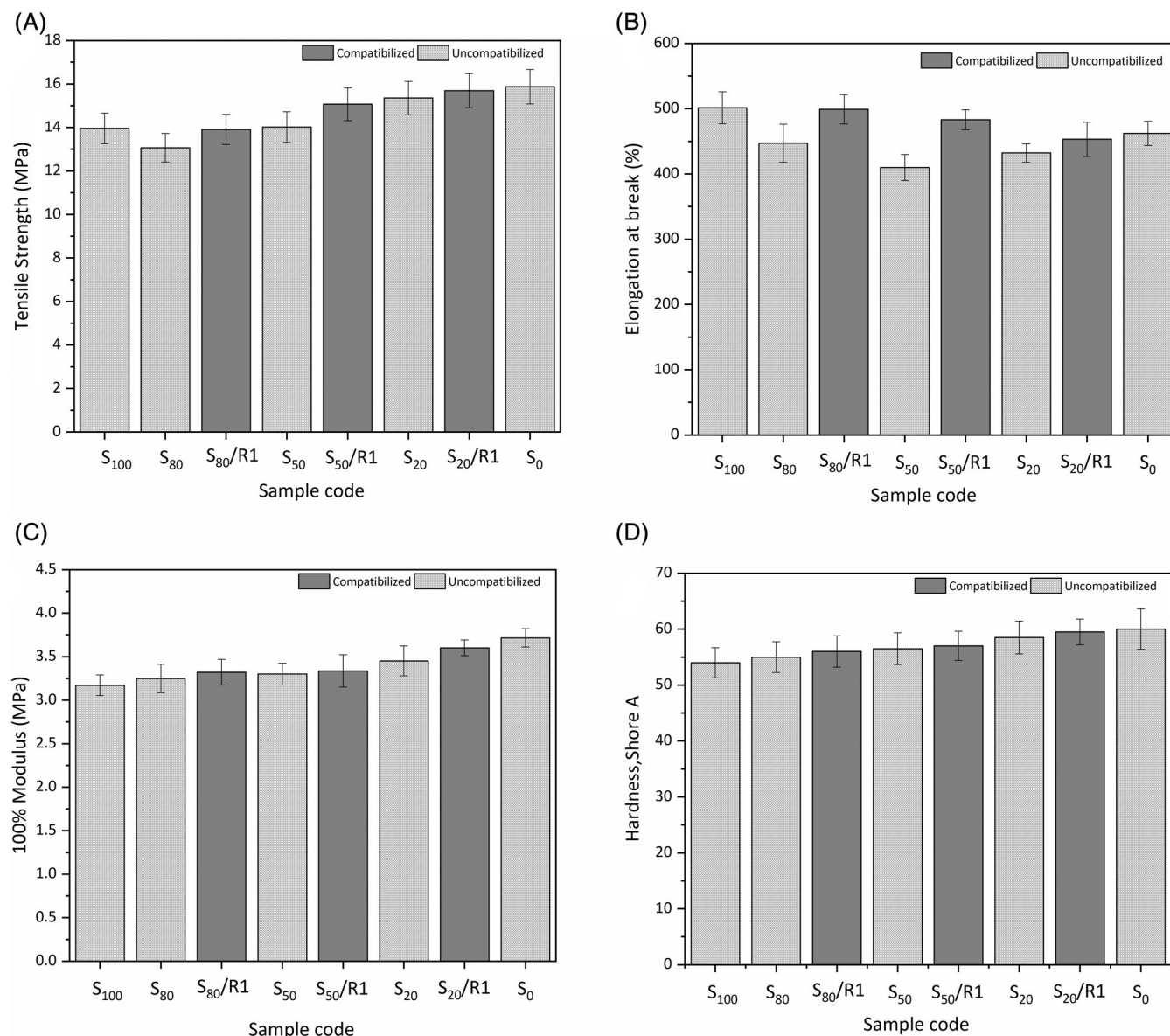


FIGURE 7 The effect of blend ratio and R1 compatibilizer. A, Tensile strength, B, Elongation at break, C, Young's modulus, and D, Hardness of NBR/SBR blend. NBR, nitrile butadiene rubber; SBR, styrene-butadiene rubber

preventing the flow and mobility of the SBR/NBR blends.⁴²

3.4.2 | Hardness

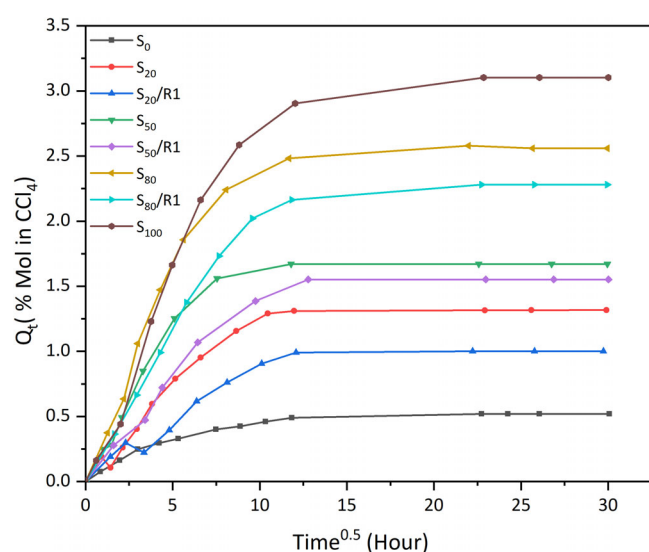
Figure 7D represents hardness (Shore A) for all blend's ratio in the presence of R1 as compatibilizer. As it manifests, presence of the compatibilizer resulted in higher hardness. This is because of an improvement in phase adhesions and higher cross-link density. As it has been showed, the presence of compatibilizer affected the mechanical behavior of the blends because all the mechanical properties were critically improved.

3.4.3 | Swelling behavior and cross-link density

Swelling behavior and effect of compatibilizer on carbon tetrachloride (CCl₄) and chloroform (CHCl₃) were studied. Table 4 represents interaction parameter, cross-link density, and equilibrium swelling. It was observed that χ values decreased with an increase in the amount of SBR when carbon tetrachloride was used and in reverse, when chloroform was used. Higher values of χ demonstrate lower polymer-solvent interactions⁴³ and the values of χ are in disagreement with Q_t values since the decrease in polymer-solvent interaction resulted in decline in the amount of solvent uptake in polymer.

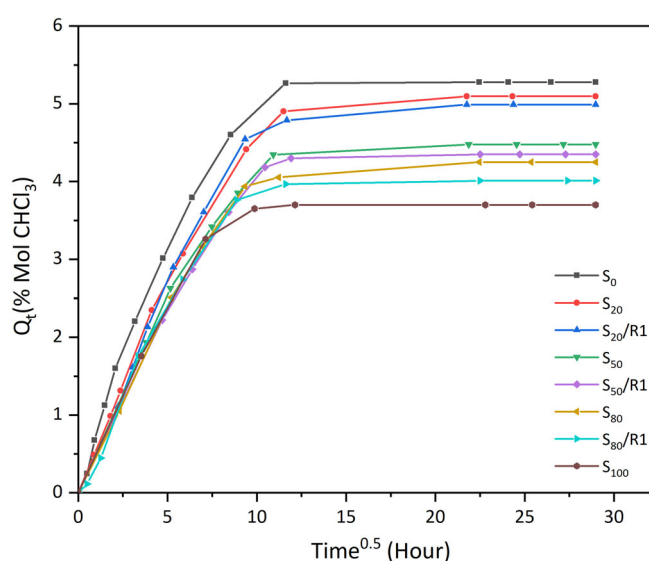
TABLE 4 Interaction parameter, cross-link density, and equilibrium swelling

Sample code	Interaction parameter, χ		Cross-link density, $\nu \times 10^{-4}$ (mol/cm ³)		Equilibrium swelling, Q_t (%)	
	CCl ₄	CHCl ₃	CCl ₄	CHCl ₃	CCl ₄	CHCl ₃
S ₀	0.48	0.15	18.19	1.12	0.55	5.27
S ₂₀	0.41	0.19	9.16	1.52	1.31	5.09
S ₂₀ /R1	0.39	0.23	10.34	2.06	1.01	4.98
S ₅₀	0.27	0.36	7.84	2.36	1.67	4.47
S ₅₀ /R1	0.33	0.39	8.52	2.86	1.55	4.35
S ₈₀	0.17	0.44	5.56	3.97	2.55	4.21
S ₈₀ /R1	0.19	0.46	6.36	4.62	2.27	4.01
S ₁₀₀	0.11	0.46	4.38	6.05	3.10	3.70

**FIGURE 8** Effect of blend composition and the presence of R1 compatibilizer on mol uptake of CCl₄ at room temperature [Color figure can be viewed at [wileyonlinelibrary.com](#)]

The calculated values of ν also are in agreement with this trend. The molecular weight between two cross-links, M_c , depends on cross-link density as $\nu = 1/(2M_c)$, and ν contains two parameters (ν_{chem} and ν_{inti}), ν_{chem} expresses cross-link density of C—S_x—C chemical bonds, and it will not change during swelling and ν_{inti} expresses cross-link density due to entanglements and crystallinity.⁴⁴ Therefore, ν_{inti} can change during swelling.^{45,46} Accordingly, the discrepancy in rubber chain entanglements density and penetration factor in different solvents could be the reason for the different ν and M_c values.

As it can be seen, the swelling is decreased in the presence of the compatibilizer, the interaction parameter increased, and equilibrium swelling decreased. This observation showed that more cross-links could be formed between the rubber chains which would increase the adhesion between blend components and reduce the free

**FIGURE 9** Effect of blend compositions and the presence of R1 compatibilizer on mol uptake of CHCl₃ at room temperature [Color figure can be viewed at [wileyonlinelibrary.com](#)]

volume between domains' components. Figure 8 shows the CCl₄ solvent sorption behavior of different blend compositions at room temperature. Solvent uptake agrees with SBR content. This issue occurs because of polarity of non-polar solvent and SBR. Figure 9 shows the same thing for CHCl₃ solvent. As it expected, CHCl₃ uptake tendency decreases with an increase in SBR content.

This is due to low interaction between SBR and a polar solvent. Interaction parameters calculated from Flory–Rehner confirms these observations.

3.5 | Morphology study

Figure 10 represents SEM images of fracture surfaces of 50/50 NBR/SBR blend without and with compatibilizer.

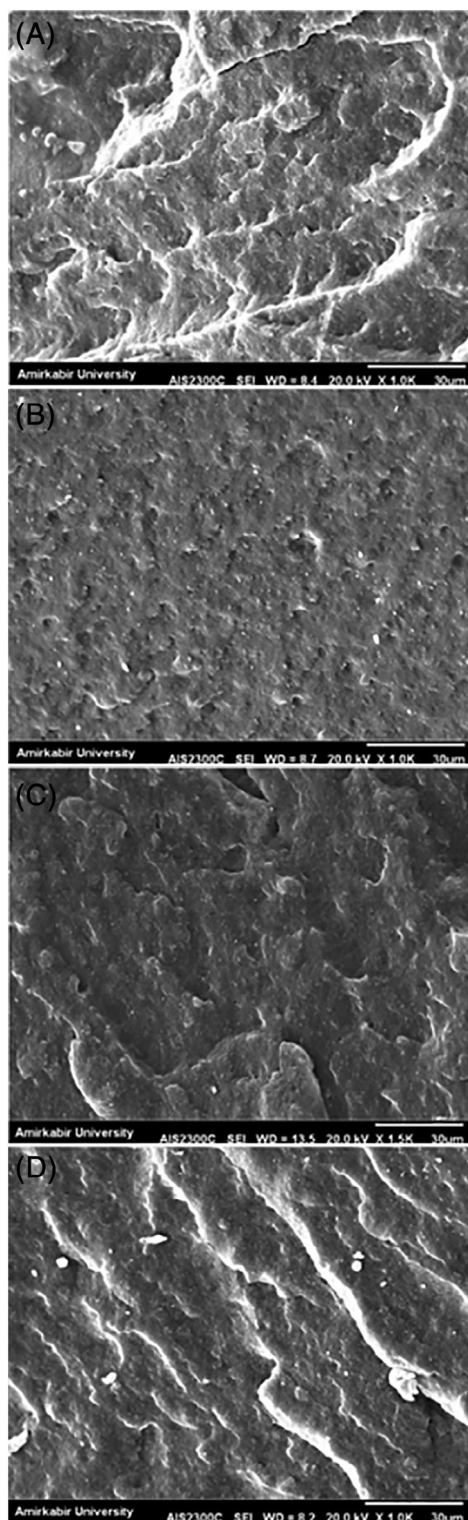


FIGURE 10 SEM micrographs showing tensile fracture surface of SBR/NBR blends 50/50. A, Without compatibilizer and B-D, With R1, R4, and R5 compatibilizers, respectively. NBR, nitrile butadiene rubber; SBR, styrene-butadiene rubber; SEM, scanning electron microscope

This morphology shows rough surface, and irregular domain shapes compared with the sample that contained grafted compatibilizer. Figure 10B-D shows (50/50)

NBR/SBR blend with R1, R4, and R5 compatibilizer respectively; as it can be seen, phases are not distinguishable. However, at the presence of R1 compatibilizer, the fracture surface of the blend is smoother and surface texture had increased significantly. This micrograph indicates the presence of R1 compatibilizer, improved homogeneity, uniformity of surface, and subsequently, the compatibility blends and confirms the DMTAs test, which was discussed earlier.

4 | CONCLUSIONS

The effect of GMA-grafted SBR in the presence of styrene comonomers as a compatibilizer agent on microstructure, compatibility, mechanical properties, swelling, and curing behavior of blend based on SBR and NBR was studied. The highest value of grafting yield was achieved when 1.5 phr GMA and 1/1 ratio of styrene/GMA were used. The optimized state resulted in this circumstance since GMA monomers reached the higher value of grafting here in addition to negligible amount of GMAs homopolymerization and also the least amount of gel made during the procedure. Higher grafting yield does not necessarily create better tendency between components as the increase in viscosity of compatibilizer and epoxy groups of GMA in SBR-g-GMA-co-St reduced the compatibility efficiency. The presence of compatibilizer caused shorter scorch time t_2 and the cure time t_{90} . Furthermore, the minimum torque, M_L , decreased; however, the maximum torque, M_H , increased in all blends' ratios. The incorporation of R1 compatibilizer also contributed to enhance the physical-mechanical properties such as: tensile properties, modulus, and elongation at break compared to uncompatibilized blends. R1 compatibilizer improved swelling behavior of SBR/NBR blends by increasing cross-link density and better interfacial adhesion of blend components which was in agreement with SEM micrographs that illustrated coarser surface at the absence of compatibilizer.

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How to cite this article: Javadzadeh M, Arefazar A, Zakizadeh M. The effect of glycidyl methacrylate and styrene comonomers on compatibility, physical-mechanical properties, and swelling behavior of SBR/NBR blends. *Polym Eng Sci.* 2020;1–13. <https://doi.org/10.1002/pen.25379>